

A Hybrid A*D3QN Framework with Prior Knowledge and Multimodal Data Fusion for USV Path Planning

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Abstract—Path planning is an essential task for the mission execution of unmanned surface vehicle (USV). However, existing advanced techniques based on deep reinforcement learning (DRL) often suffer from low learning efficiency and insufficient environmental perception from single-sensor configurations. To address these issues, this paper proposes a hybrid framework named A*D3QN, which integrates the heuristic efficiency of the A* algorithm with the adaptive decision-making of a dueling double deep Q-network (D3QN), associated with the multimodal data fusion for precise environment modeling. The proposed A*D3QN incorporates prior knowledge from the global paths generated by A* to initialize and guide the D3QN learning process. The prior knowledge, formatted as RL transition tuples, is used in the reward function design and the N-step prioritized experience replay, which significantly accelerates overall learning efficiency. Moreover, an improved D3QN architecture is designed to dynamically fuse visual data and navigation states via a cross-entropy attention mechanism, enabling multimodal perception in partially unknown environments. Extensive experiments across three scenarios with varying obstacle densities demonstrate that A*D3QN significantly outperforms state-of-the-art DRL baselines. Ablation studies further validate the necessity of each component.

I. Introduction

The unmanned surface vehicle (USV) plays an important role in various maritime domains, such as oceanographic mapping, safety surveillance, and military operations. To ensure that the USV can perform missions safely and effectively, accurate path planning must be guaranteed. However, real-world maritime environments are generally characterized by uncertainty and dynamics, such as submerged reefs and floating buoys. Lack of complete prior information severely limits the potential for USV path planning in critical applications [1].

Considerable research has been conducted on the path planning task. Numerous algorithms have been proposed to achieve remarkable efficiency in generating end-to-end trajectories within well-defined and static environments. For example, the heuristic A* algorithm [2] has been extensively utilized in path planning due to its adaptable and effective traits, while evolutionary algorithms such as neural population dynamics [3] and

alpha evolution [4] have further improved the planning efficiency. However, these algorithms are mainly designed for static configurations, which generally lack flexibility. Once the environment changes, the visibility map must be reconstructed, accordingly paths re-planned, resulting in mission completion with mixed success.

Deep reinforcement learning (DRL) has emerged as a sophisticated choice for autonomous path planning, enabling agents to learn optimal strategies through direct mappings from environmental states to actions. This capability is particularly critical in real-time marine environments, where dynamic obstacles and partial observability demand adaptive decision-making [5], [6]. Among numerous DRL studies, deep Q-network (DQN) and its variants, such as double DQN (DDQN) and dueling DDQN (D3QN), have dominated recent state-of-the-art research efforts [7], [8]. For instance, NPDDQN [9] integrated DDQN with an experience screening mechanism to enhance obstacle avoidance. Some work incorporated domain-specific knowledge [10] or global heuristics to accelerate policy learning and improve path planning efficiency [11]. G2RL [12] enhances DQN's policy learning efficiency by integrating A* algorithm's global guidance through reward function design. Despite these advances, two critical limitations remain in current DRL-based path planning. 1) Low learning efficiency: Most DQN variants rely on extensive environmental interactions to converge to optimal policies, leading to slow training speeds and high computational costs. 2) Insufficient environmental perception: Existing frameworks mainly utilize navigation variables, i.e., position, speed, and heading angle, as state representations, neglecting multimodal sensor data such as visual input from cameras that could enrich environmental understanding. This oversight limits the agent's ability to perceive complex maritime scenarios, such as distinguishing underwater obstacles or adapting to dynamic environmental changes. A robust fusion of multimodal sensory inputs is essential to bridge this gap and enhance decision-making reliability in real-world applications.

To address the aforementioned challenges, this paper proposes a novel USV path planning algorithm named A*D3QN. It combines the static, yet computationally efficient, A* algorithm with the recent D3QN framework to achieve autonomous obstacle avoidance and efficient path planning in partially unknown environments. The proposed algorithm consists of two main inspirations.

- To effectively reduce redundant exploration and

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accelerate RL convergence, the proposed A*D3QN framework integrates the A* algorithm to generate global obstacles-aware paths based on known environmental information. The generated paths are formatted as RL transition tuples, which are considered as prior knowledge to accelerate the D3QN learning process. In the proposed A*D3QN, a hybrid reward function is designed by integrating the factors of global path guidance, obstacle avoidance, repeated exploration penalties, and goal achievement as a whole, which provides illuminating knowledge to encourage the use of prior knowledge in USV path planning. Moreover, the generated RL transition is also saved in the replay buffer of an N-step prioritized experience replay mechanism, where learning pressure is enforced to balance the short- and long-term rewards.

- To realize precise environmental perception through multimodal sensor configurations, an improved D3QN architecture is designed. It provides a new multimodal information processing and fusion strategy to combine visual data and navigation states for fast and accurate environment modeling of USV path planning.

II. USV Path Planning Based on A*D3QN

In this section, the details of A*D3QN are described, including the construction of prior knowledge by A* global path planning and the improved D3QN architecture. The algorithmic description is shown in Algorithm 1.

A. Global Path Planning with A* Algorithm

To boost the real-time path planning of RL, the proposed algorithm first leverages prior map information by running the A* algorithm to compute an available route from the starting point to the destination. A* evaluates the priority of each position according to a heuristic function

$$f(n) = g(n) + h(n), \quad (1)$$

where $f(n)$ is the comprehensive cost, $g(n)$ is the cost of distance that USV has traveled from the start position to the current grid, and $h(n)$ is the estimated cost of distance from the target position.

Given a static environment, A* is capable of quickly learning the global path, denoted as P_{glob} , which is further transformed into RL transition tuples, that is, in the form of (s_t, a_t, r_t, s_{t+1}) , and stored in the replay buffer. These stored transitions, together with the global path, are treated as prior knowledge to be further used for the initialization, reward function design, and experience replay of D3QN. The above process is described on lines 3-10 of Algorithm 1.

Algorithm 1 A*D3QN Algorithm

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1: Initialize episodes  $E$ , learning rate  $\alpha$ , discount factor  $\gamma$ , N-step  $n$ , replay buffer  $\mathcal{D}$ , exploration rate  $\epsilon$ , maximum steps per episode  $T$ , and a small positive number  $\epsilon'$ ; Randomly initialize current network parameters  $\theta$  and target parameters  $\theta^-$ .
2: Initialize start position  $(x_s, y_s)$  and end position  $(x_g, y_g)$ .
3: Use A* to plan global path  $P_{\text{glob}}$ .
4: for each step  $(s_t, a_t)$  in  $P_{\text{glob}}$  do
5:   Obtain  $s_t$ , select  $a_t$  using  $\epsilon$ -greedy policy.
6:   Execute  $a_t$ , obtain N-step reward  $r_t$  and  $s_{t+1}$ .
7:   Calculate TD error  $\delta_t = G_t^{(n)} - Q(s_t, a_t; \theta)$ .
8:   Compute priority  $p_t = |\delta_t| + \epsilon'$ .
9:   Store transition  $(s_t, a_t, r_t, s_{t+1})$  with  $p_t$  in  $\mathcal{D}$ .
10: end for
11: for  $e = 1$  to  $E$  do
12:   if  $e \leq 10$  then
13:     Sample a batch of experiences from  $\mathcal{D}$ .
14:     Update current network parameters  $\theta$ .
15:     At intervals update target parameters  $\theta^- \leftarrow \theta$ 
16:   else
17:     Initialize environment and state  $s_0 = \{S_1, S_2\}$ .
18:     for  $t = 1$  to  $T$  do
19:       Obtain  $s_t$ , select  $a_t$  using  $\epsilon$ -greedy policy.
20:       Execute  $a_t$ , obtain N-step reward  $G_t^{(n)}$  and  $s_{t+1}$ .
21:       Calculate TD error  $\delta_t = G_t^{(n)} - Q(s_t, a_t; \theta)$ .
22:       Compute priority  $p_t = |\delta_t| + \epsilon'$ .
23:       Store transition  $(s_t, a_t, r_t, s_{t+1})$  with  $p_t$  in  $\mathcal{D}$ .
24:       if  $\text{size}(\mathcal{D}) \geq \text{batch size}$  then
25:         Sample a batch of experiences from  $\mathcal{D}$ .
26:         Update current network parameters  $\theta$ .
27:         At intervals update target parameters  $\theta^- \leftarrow \theta$ .
28:       end if
29:     end for
30:   end if
31: end for

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B. Real-Time Path Planning with Improved D3QN

In order to realize precise environmental perception, multimodal sensors are configured, where visual and navigation data are taken simultaneously into account for path planning. Together with the integration of prior knowledge, an improved D3QN architecture is proposed, where its graphical illustration is shown in Figure 1.

1) State Space Design: The environmental state represents the information locally controlled by the agent at time step t . It consists of two components: the visual state variables S_1 and the navigation state variables S_2 . The visual state S_1 is initially represented as a grayscale image that can originate from previously collected sensor data and can be transformed into a color image by overlaying a navigation information channel instead of

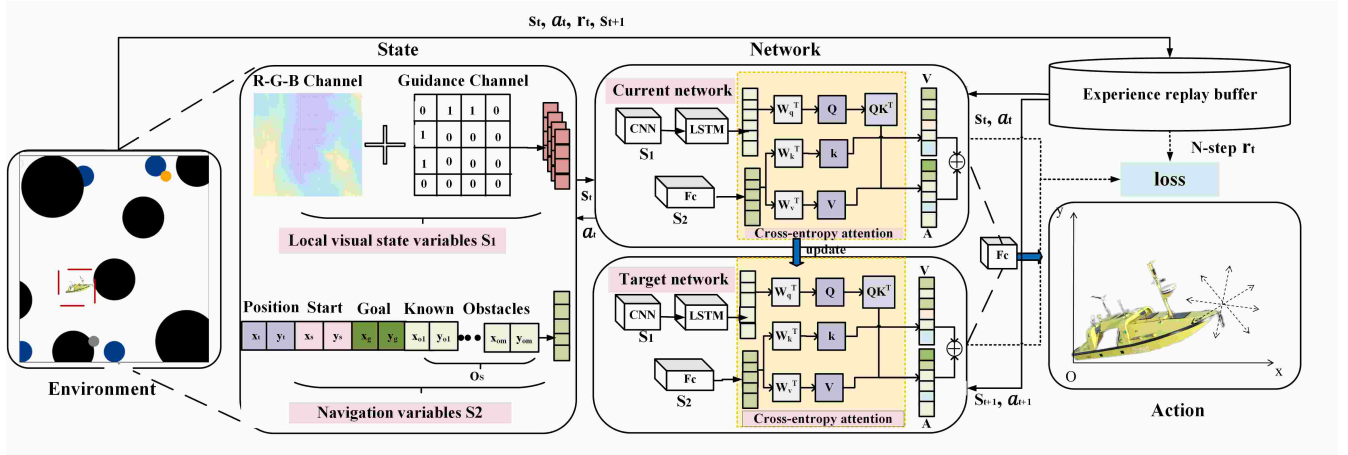


Fig. 1. The overall framework of the improved D3QN

obstacles. Thus, the visual state is represented as a multi-channel tensor $S_1 \in \mathbb{R}^{4 \times 4 \times 4}$. The navigation state S_2 directly encodes position using coordinate values, including current position (x_t, y_t) , start position (x_s, y_s) , goal position (x_g, y_g) , and set of known static obstacles $O_s = \{(x_{o1}, y_{o1}), (x_{o2}, y_{o2}), \dots, (x_{om}, y_{om})\}$.

2) Action Space Design: For each state, the agent needs to choose an action from action space $A = \{a_1, a_2, \dots, a_N\}$. In this study, the USV is capable of executing discrete actions among eight moving directions: forward, backward, left, right, forward-left, backward-left, forward-right, and backward-right, and each action above is assigned a unique numerical identifier ranging from 0 to 7. This numerical encoding enables to facilitate pose transformation and integration with the neural network.

3) Reward Function Design: In the D3QN algorithm, the reward function is one of the most critical factors to guide the USV to achieve efficient and safe navigation. It is composed of five parts: distance-based reward, collision penalty, guiding reward, exploration penalty, and achievement reward.

Distance-based reward (R_1): This component incentivizes the USV to minimize its distance from the target location.

$$R_1 = k_1 \sqrt{(x_t - x_g)^2 + (y_t - y_g)^2}, \quad (2)$$

where (x_t, y_t) is the position of the USV at time step t , (x_g, y_g) is the distance between the USV and the target position.

Collision penalty (R_2): To discourage proximity to obstacles, a penalty is applied when the USV is within a critical distance threshold.

$$R_2 = \begin{cases} k_2 & d_{\min} \leq 0.5, \\ 0 & d_{\min} > 0.5, \end{cases} \quad (3)$$

where $d_{\min}$ is the closest distance between the USV and obstacles.

Guiding reward (R_3): This reward encourages RL to generate paths that are consistent with the prior global path obtained by A^* .

$$R_3 = \begin{cases} k_3 & \text{if } P_{\text{current}} \in P_{\text{glob}}, \\ 0 & \text{otherwise,} \end{cases} \quad (4)$$

where P_{current} is the current position of USV. R_3 is intended to strengthen RL exploitation via prior knowledge obtained by A^* .

Exploration penalty (R_4): To promote efficient exploration and prevent redundant path rewards, a penalty is imposed for revising previously explored states.

$$R_4 = \begin{cases} k_4 & \text{if } P_{\text{current}} \in P_{\text{hist}}, \\ 0 & \text{otherwise,} \end{cases} \quad (5)$$

where P_{hist} denotes the historical path of the current episode.

Achievement reward (R_5): The achievement reward is granted once the path planning task has been completed successfully within a predefined maximum step, denoted R_5 .

Consequently, the comprehensive reward function is designed as follows.

$$r = \begin{cases} R_5 & \text{if task completed,} \\ R_1 + R_2 + R_3 + R_4 & \text{if otherwise.} \end{cases} \quad (6)$$

4) Improved Network Architecture: In the proposed framework, D3QN is adopted as the base learner for RL. As a recent state-of-the-art variant of DQN, D3QN integrates double DQN and dueling DQN to simultaneously address the issues of overestimation and inefficient Q-value learning [13]. The target Q-value is computed as

$$y^{\text{D3QN}} = r_t + \gamma \left[V(s_{t+1}) + A \left(s_{t+1}, \arg \max_{a \in A} Q(s_{t+1}, a; \theta); \theta^- \right) - \frac{1}{|A|} \sum_{a \in A} A(s_{t+1}, a; \theta^-) \right], \quad (7)$$

where $|A|$ is the scale of action space, $V(s_t)$ is the state value function, and $A(s_{t+1}, a)$ denotes the advantage of action a . γ is the discount factor. θ and θ^- are the parameters of current and target networks, respectively.

As mentioned above, this paper proposes an improved D3QN network architecture that fuses visual and navigation data to achieve precise environmental perception, as shown in Figure 1. Specifically, CNN and LSTM networks are used to extract features from visual variables S_1 , while navigation variables S_2 are handled using fully connected layers, which are essentially distinguished from original D3QN architecture. Cross-entropy attention [14] is an effective mechanism in multimodal tasks where information from different modalities needs to be integrated to solve a specific problem. In this work, the technique is employed to facilitate feature interaction and fusion between the two modalities. The detailed architecture is illustrated in Figure 1 (see cross-entropy attention component). This design allows for dynamic alignment of visual and spatial features, thereby improving the learning efficiency and robustness of D3QN.

5) N-Step Prioritized Experience Replay: Prioritized Experience Replay (PER) [15] is an advanced experience replay mechanism that strategically selects transitions from the replay buffer $\mathcal{D}$ based on their significance of Temporal Difference (TD) errors. In this paper, we employ the N-step variant of prioritized experience replay, which incorporates multi-step returns to update the Q-value based on a sequence of future rewards. The return of n -step at current time step t is denoted as $G_t^{(n)}$, which is defined as follows.

$$G_t^{(n)} = \sum_{i=0}^{n-1} \gamma^i r_{t+i+1} + \gamma^n \max Q(s_{t+n}, a'; \theta^-), \quad (8)$$

where a' denotes the action under state s_{t+n} . Then, TD error δ_t is calculated by

$$\delta_t = G_t^{(n)} - Q(s_t, a; \theta). \quad (9)$$

This study adopts a SumTree structure to store samples in prioritized experience replay. Each transition sample, along with its priority, is stored in the leaf nodes of the tree. The transition sample priority is defined as

$$p_t = |\delta_t| + \epsilon', \quad (10)$$

where ϵ' is a small constant to ensure stability.

III. Experimental Studies

In this section, we perform simulation experiments in different partially known environments. The proposed A*D3QN is compared with A* and recent state-of-the-art DRL algorithms for performance verification.

A. Experimental Environment and Parameter Settings

Considering the navigation characteristics of the USV in marine environments, we construct a 2-D simulation environment using Python 3.7. The environment comprises a 20×20 grid map, corresponding to an actual area of $20\text{m} \times 20\text{m}$. Three experimental scenarios are established with known-to-unknown obstacle ratios of 3:1, 1:3, and 1:5, denoted as Scenario 1, Scenario 2, and Scenario 3, respectively. In this paper, obstacles are randomly generated and represented using enveloped circles, which more accurately reflect real-world conditions [16]. Known obstacles are colored dark blue, and unknown obstacles are colored black. The training parameters for the A*D3QN algorithm are set as follows: a total of 1500 training episodes, a maximum of 200 steps per episode, and a vehicle speed of 1 m/s in still water. The learning rate is fixed at 0.0001, and n of the N-step is set to 5. An ϵ -greedy policy is employed to balance exploration and exploitation. The current network and the target network are updated every 30 episodes. All paths are smoothed by the B-spline to improve continuity. In our experiments, the parameters of the reward function are set as follows: $k_1 = 0.05$, $k_2 = -1$, $k_3 = -0.02$, $k_4 = 0.2$, and $k_5 = 1$.

B. Simulation and Analysis

This section conducts extensive experiments to verify obstacle avoidance and path planning abilities, as well as convergence speed.

1) Comparative Experiments with A* Algorithm: The results of path planning are shown in Figure 2. It can be observed that A*D3QN demonstrates superior adaptability and path planning ability compared to A*. In Scenario 1 and Scenario 2, A*D3QN can generate paths comparable to or even shorter than those of A*. In Scenario 3, when the obstacles are more unknown, A* fails to complete the task due to lack of foresight. In contrast, A*D3QN can proactively avoid potential obstacles and generate robust paths.

2) Comparison experiment with DRL algorithms: In order to verify the real-time path planning, the proposed algorithm is compared to two recent state-of-the-art DRL variants, namely G2RL and NPDDQN. The experimental results are shown in Figures 3, 4, and Table I. In Scenario 3, G2RL fails to acquire a policy capable of reaching the designated goal. As a result, its performance metrics are omitted from Table I. The final test path of G2RL in scenario 3 is shown in Figure 3, where it can be observed that G2RL collided with an obstacle. As shown in Table I, A*D3QN demonstrates comprehensive advantages in all three scenarios. In terms of the efficiency of path planning, the average path lengths are 23.38m, 18.90m, and 26.73m, respectively, which are superior to the NPDDQN and G2RL algorithms. A*D3QN maintained a higher success rate in three cases, which are 90%, 93% and 89%, respectively. As shown in Figure 4, the number of successful episodes of A*D3QN

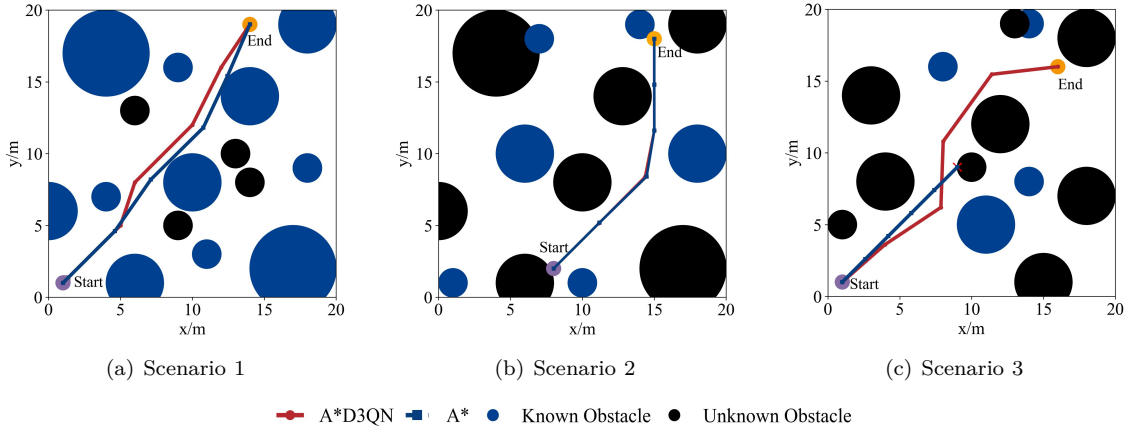


Fig. 2. Comparison of paths planned with A* algorithm.

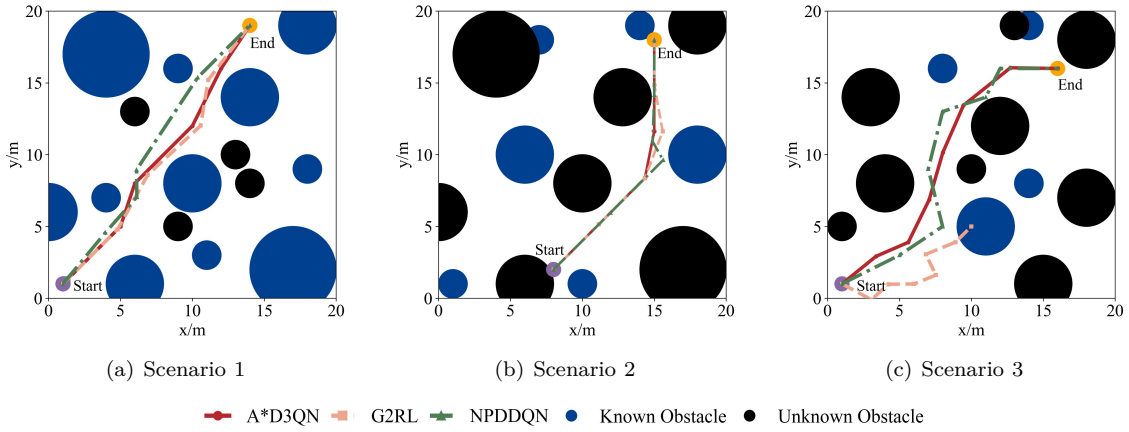


Fig. 3. Comparison of paths planned with G2RL and NPDDQN.

TABLE I
Experimental results in tree scenarios

Scenario	Algorithm	Minimum length(m)	Average length(m)	Success rate(%)
Scenario 1	G2RL	29.38	39.46	86
	NPDDQN	31.63	42.56	63
	A*D3QN	23.38	32.06	90
Scenario 2	G2RL	20.56	27.34	76
	NPDDQN	21.73	26.69	84
	A*D3QN	18.90	25.69	93
Scenario 3	NPDDQN	32.97	35.69	84
	A*D3QN	26.73	29.35	89

is always high, with the fastest growth rate among the three scenarios, and it is close to linear. This high and steadily increasing success rate also indirectly reflects the convergence speed. Therefore, the results further indicate that the convergence speed of A*D3QN is significantly faster.

The heuristic guidance of the A* algorithm can improve the stability, which leads to faster convergence rates of A*D3QN and G2RL in Scenario 2. However, with increasing unknown information, the guiding role of A* will be diminished, and more autonomous DRL learning

will be needed to avoid obstacles. The performance of DQN is often erratic due to the inaccuracy of Q-value estimation and insensitivity to environmental feedback. Therefore, G2RL gets worse and worse in scenarios 1 and 2. A*D3QN and NPDDQN improve the learning efficiency by obtaining higher quality experience samples in N-step prioritized experience replay. However, NPDDQN exhibits the lowest successful episodes due to lack of prior knowledge.

3) Ablation Studies: To fully verify the effectiveness of the proposed method, a series of ablation experiments were conducted. The ablation comparison algorithms are shown in Table II and the results of the average rewards and convergence episodes are presented in Figure 5. The findings show that A*D3QN achieves the highest average reward and the fastest convergence. A*D3QN-A converges the slowest due to A* being removed. The results show that embedding global path information is crucial for guiding exploration. A*D3QN outperforms A*D3QN-C. This is because A*D3QN-C removes the visual data, which makes its state characteristics incomplete. Especially in the real environment, when there is noise in the data of a single sensor, the experimental

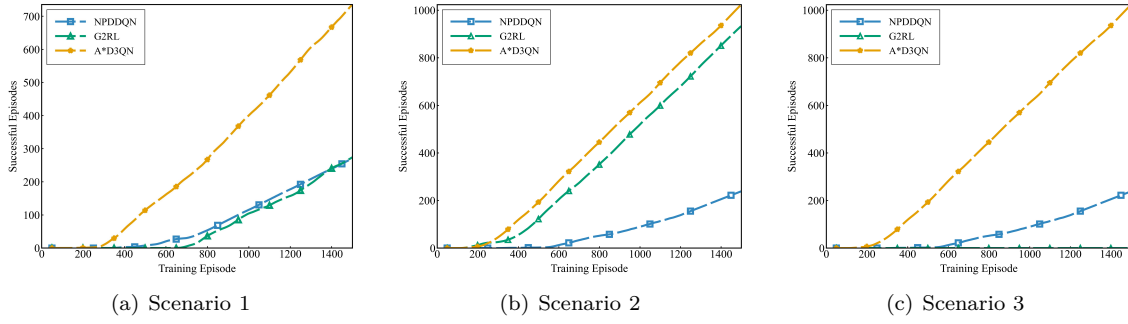


Fig. 4. Comparison of the episodes reaching the goal.

results may be worse. Interestingly, the average reward of A*D3QN-C is lower than that of A*D3QN-X. This observation highlights the importance of the cross-entropy attention mechanism. A*D3QN-N, which removes the N-step mechanism, significantly reduces DRL's performance. The N-step mechanism can optimize performance through effective bias-variance trade-off management between immediate and long-term returns.

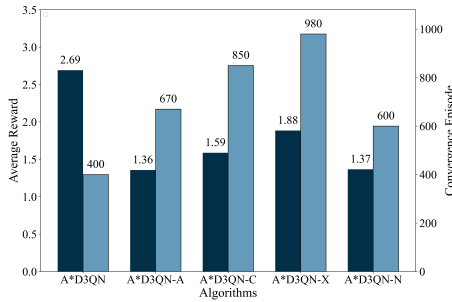


Fig. 5. Ablation experiment results.

To further evaluate the performance of A*D3QN, scenario 2 is divided into two extremes: fully known and completely unknown. The reward curves are shown in Figure 6. The experimental results indicate that prior knowledge has a certain impact on the convergence speed. Nevertheless, A*D3QN maintains stable and efficient convergence across all three environments. This is attributed to the effective integration of the global guidance A* and the real-time adaptability of D3QN.

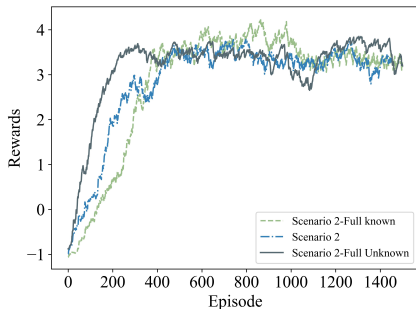


Fig. 6. Rewards of A*D3QN algorithm.

IV. CONCLUSIONS

This study introduces A*D3QN, a hybrid real-time path planning framework designed for USV in partially known maritime environments. The algorithm employs the A* algorithm to quickly plan a safe path that circumvents known obstacles, while simultaneously guiding the D3QN for autonomous real-time navigation. By embedding A* prior knowledge, incorporating the N-step prioritized experience replay, and the network architecture of D3QN to fuse visual state variables and navigation variables, A*D3QN enhances the learning efficiency and planning robustness. Experiments across three scenarios show that A*D3QN achieved higher success rates, shorter average path lengths, and faster convergence than NPDDQN and G2RL. Importantly, while maintaining path efficiency comparable to A*, A*D3QN offers the added benefits of adaptivity and real-time responsiveness. Ablation studies further confirmed the effectiveness of each component in A*D3QN in boosting learning performance and convergence stability. However, the proposed A*D3QN uses discrete actions and does not model USV dynamics. Additionally, the algorithm's performance could be affected by sensor noise in actual marine environments. Future work will incorporate continuous control and dynamic modeling to improve real-world applicability while improving robustness to sensor noise.

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TABLE II
Ablation Comparison Algorithms

Algorithm	A*	S_1	S_2	Cross-entropy Attention	N-Step
A*D3QN-A	✗	✓	✓	✓	✓
A*D3QN-C	✓	✓	✓	✗	✓
A*D3QN-X	✓	✗	✓	✓	✓
A*D3QN-N	✓	✓	✓	✓	✗
A*D3QN	✓	✓	✓	✓	✓

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